

Strategic Intelligence Briefing: Analysis of the 22.2 Tbps Hyper-Volumetric DDoS Attack (September 2025 Case Study)

I. Executive Summary: The New Paradigm of Digital Siege (2025)

The Distributed Denial-of-Service (DDoS) landscape in 2025 has undergone a profound and immediate escalation, moving from frequent disruption to warfare defined by unprecedented scale. The successful mitigation of a record-shattering 22.2 Terabits per second (Tbps) attack in September 2025 serves as a critical strategic case study, demanding an urgent reassessment of defense capacities across all critical infrastructure sectors. This event confirms that the era of "hyper-volumetric" threats—attacks designed to overwhelm networks with previously unimaginable scales of traffic—is now standard operational reality.

1.1 Contextualizing Hyper-Volumetric Threats (2025)

The rate of attack growth leading up to this record has been exponential, signaling a fundamental paradigm shift in digital conflict intensity. In the first quarter of 2025 alone, a major mitigation provider blocked 20.5 million DDoS attacks, a figure nearly matching 96% of the entire volume blocked throughout the calendar year 2024.¹ This staggering increase confirms that the overall volume and frequency of assaults have reached a threshold that renders manual, human-driven security processes obsolete. The only sustainable response to this velocity and scale is fully automated, machine-speed mitigation capacity.³

The definition of a massive cyber assault has been permanently redefined by the emergence of hyper-volumetric campaigns.⁵ In the second quarter of 2025, over 6,500 hyper-volumetric DDoS attacks—those exceeding 1 Tbps or 1 Billion packets per second (Bpps)—were mitigated, averaging approximately 71 such incidents per day.⁶ This level of continuous,

high-intensity assault confirms that multi-terabit threats are an active and ongoing category of risk, not just theoretical anomalies. Furthermore, Layer 3/Layer 4 (L3/L4) Network-layer attacks continued to dominate the threat landscape in volume, recording a drastic increase of 397% Quarter-over-Quarter (QoQ) and a 509% Year-over-Year (YoY) surge in Q1 2025, underscoring the necessity of high-capacity volumetric defense tailored specifically for network saturation.²

1.2 Case Study Snapshot: Key Metrics of the 22.2 Tbps Flash Assault

This incident represents the apex of threat actor capability observed thus far, deliberately orchestrated to maximize network disruption within the shortest possible timeframe.

Metric	Detail	Contextual Significance
Attack Date	Late September 2025 ⁴	Confirms rapid escalation following earlier 2025 records.
Peak Scale (Volume)	22.2 Terabits per second (Tbps) ⁴	Double the size of the previous record (11.5 Tbps).
Peak Scale (Packet Rate)	10.6 Billion packets per second (Bpps) ⁸	Extreme packet intensity designed to exhaust network hardware state tables.
Duration	A rapid, "hit-and-run" style assault lasting only 40 seconds ³	Intentionally short, demanding instantaneous, autonomous mitigation.
Target	Single IP address of an unnamed European network infrastructure company ⁸	High-value, critical infrastructure target ensuring maximum downstream collateral

		impact.
Technology	UDP flood utilizing a unique " carpet bomb " port distribution technique ⁸	Network layer attack exploiting packet-intensive protocols to evade traditional perimeter defenses.
Suspected Source	The Aisuru IoT botnet , comprising over 404,000 unique, non-spoofed source IPs ⁸	Highlights the pervasive risk of compromised Internet-of-Things (IoT) devices powering global cyber warfare.

The rapid progression of volumetric records throughout 2025—from 7.3 Tbps to 11.5 Tbps, culminating in the 22.2 Tbps peak ⁹—strongly suggests an active, deliberate campaign by sophisticated threat actors to continuously map and stress-test the scrubbing capacity of major cloud protection providers. This systematic probing behavior goes beyond simple disruption; it is a critical step in intelligence gathering, preparing for future, highly targeted supply-chain operations that aim to exploit vulnerabilities revealed by the mitigation response time.

II. The Target and Operational Chronology

The selection of the target and the velocity of the attack were strategically designed to maximize potential damage and challenge the resilience of modern network defenses.

2.1 Incident Identification and Timeline

The 22.2 Tbps assault was characteristic of the "flash" DDoS tactic, executed as a "hit-and-run" campaign lasting only 40 seconds.³ This short duration is a key operational choice, as it is precisely engineered to inflict maximum strain on intermediate network forwarding components, such as edge routers and firewall appliances, before traditional detection and manual mitigation processes can be initiated.⁴ The objective is to trigger critical

failures and network instability in the brief window before automated cloud defenses can absorb the full load.

This attack continued an intense sequence of record-breaking volumetric events observed throughout the year. The 22.2 Tbps incident successfully doubled the size of the previous 11.5 Tbps record ⁸, which had itself been established only weeks earlier, surpassing the 7.3 Tbps record set in May 2025.⁹ This accelerated timeline underscores the current volatility of the threat landscape and the continuous efforts by botnet operators to leverage increasing global bandwidth capacity.

2.2 Target Profiling: Targeting Network Infrastructure

The victim of this record-breaking assault was a single IP address associated with an unnamed European network infrastructure company.⁸ The strategic significance of targeting infrastructure and service providers cannot be overstated. Data from 2025 Q2 indicates that the Telecommunications, Service Providers, and Carriers sector consistently ranks among the most targeted industries globally.⁶

The choice of target confirms that threat actors prioritize infrastructure components. Successfully disrupting foundational service providers ensures maximum collateral damage, affecting numerous downstream clients, including commercial enterprises, government agencies, and consumer services. This intentional focus on high-impact sectors mandates that network architects recognize the hyper-volumetric scale utilized—a 22.2 Tbps flood—is necessary for the attackers because the underlying infrastructure they are attacking possesses robust, high-capacity links and distributed redundancy. The massive resource expenditure required by the 400,000-node botnet ⁸ confirms the high value placed on crippling these foundational internet services.

2.3 Weaponization of Bpps: A Greater Threat

Security architects must recognize that the packet rate metric is often a more accurate indicator of catastrophic network instability than pure volumetric flow. The concurrent peak of **10.6 Bpps** in the 40-second assault represents a profound threat to network stability.³

High packet rates stress stateless network equipment—specifically edge routers and specialized scrubbing hardware—forcing them to process an unmanageable number of individual packets and connection attempts per second. This rapid resource exhaustion,

sometimes leading to device kernel panics or forced reboots, often results in system failure long before the network's total bandwidth capacity is saturated. The attackers are intentionally targeting the processing limits of high-performance forwarding components, effectively using packet density as a weapon to induce hardware failure.¹²

III. Technical Dissection: The Hyper-Volumetric Vector

The methodology employed in the 22.2 Tbps attack was a highly optimized, dual-pronged approach targeting both bandwidth saturation and network hardware exhaustion simultaneously.

3.1 Scale and Vector Analysis (Layer 3/Layer 4)

The assault was primarily an extremely large-scale Layer 3/Layer 4 (L3/L4) User Datagram Protocol (UDP) flood.³ UDP floods are inherently volumetric, utilizing the stateless nature of the protocol to send large numbers of packets aimed at the target's ports. This technique consumes network bandwidth and forces the victim system to expend significant processing power handling the massive, unauthorized packet flow.¹³ Analysis confirmed that the traffic was not only **volumetrically extreme** (Tbps) but also **highly packet-intensive** (Bpps).⁴

The sheer magnitude of 10.6 Bpps necessitates that the defensive posture be capable of identifying and mitigating malicious traffic in milliseconds across all points of presence globally. The only infrastructure capable of handling this requirement is a globally distributed, massively scaled cloud architecture that utilizes autonomous detection, as demonstrated in the successful mitigation of this particular incident.⁴

3.2 Vector Analysis: The UDP Carpet Bombing Technique

The most sophisticated technical element of this assault was its use of a **"UDP carpet bomb"** technique.⁸ In a traditional UDP flood, an attacker often concentrates traffic on a few well-known ports. The carpet bomb, conversely, is a tactic designed explicitly to defeat traditional perimeter defenses.

The attack deliberately spread its traffic across a massive range of targets, aiming at an average of **31,000 destination ports per second** on the single victim IP address, peaking at **47,000 ports**.⁸

This massive port distribution mechanism is a clear evolution in L3/L4 attack tactics, revealing that threat actors are actively developing methods to circumvent the processing limitations of stateful enterprise network appliances, such as traditional firewalls and Intrusion Prevention Systems (IPS). By spreading the traffic across nearly all available ports, the attacker ensures that every part of the network stack is stressed simultaneously, accelerating the onset of resource exhaustion far faster than simple pure volumetric saturation would achieve.¹² The success of this tactic compels organizations to rely exclusively on high-capacity, stateless filtering and edge-delivered mitigation techniques.

3.3 Botnet Origin, Architecture, and Non-Spoofing Confirmation

The technical analysis of the traffic revealed critical details about the underlying threat actor's resources and operational capacity. The attack was traced to over **404,000 unique source IP addresses** distributed across 14 Autonomous System Numbers (ASNs) worldwide.⁸

A crucial finding was the confirmation that the source IPs **were not spoofed**.⁸ This fact establishes that the traffic originated directly from compromised endpoints under the attacker's command—a vast, globally orchestrated botnet—rather than being reflected traffic from innocent third-party servers (which relies on IP spoofing).

This botnet is strongly linked to the notorious **Aisuru botnet**.⁸ Aisuru specializes in exploiting vulnerabilities (both known and zero-day) in Internet-of-Things (IoT) devices, including routers (such as Totolink, Zyxel, D-Link, and others), DVRs, and IP cameras.¹⁴ The botnet is estimated to control more than 300,000 compromised nodes globally.¹⁶ The 404,000 unique, non-spoofed IPs utilized in the 22.2 Tbps attack underscore the central problem of global IoT security hygiene, confirming that every record-breaking DDoS event in 2025 is fundamentally enabled by a massive, ongoing security failure in consumer and manufacturer security practices regarding connected devices.¹⁵ This transforms the DDoS problem from a simple bandwidth battle into a critical supply chain security crisis.

IV. Attribution, Motivation, and Threat Actor Analysis

Understanding the actor behind the 22.2 Tbps assault and their strategic intent is essential for forecasting and preparing for future waves of hyper-volumetric campaigns.

4.1 The Aisuru Botnet Profile and Capabilities

The Aisuru botnet, the suspected force behind the September 2025 attack, has been active for more than a year, consistently demonstrating the ability to deliver multi-terabit assaults. It was previously linked to a massive 6.3 Tbps attack earlier in the year.⁸

Aisuru's core capability lies in its exceptional orchestration skills, managing hundreds of thousands of geographically distributed IoT nodes to launch synchronized, high-intensity, short-burst campaigns.³ The botnet relies on exploiting poor security configurations in consumer electronics.¹⁵

Furthermore, subsequent threat intelligence suggests that Aisuru operators are actively diversifying their attack vectors and protocols. Unconfirmed reports surfaced in October 2025 linking the botnet to a suspected **29.69 Tbps TCP-based carpet bomb attack** targeting major gaming platforms.¹⁶ This rapid pivot from UDP to TCP-based carpet bombing confirms the botnet's high degree of operational flexibility and its capability to exploit multiple network protocols. This requires defensive architectures to be ready for omni-protocol defense readiness, emphasizing that security leaders must assume that any hyper-volumetric attack faced is part of a larger, composite threat operation, often involving secondary L7 probes for weakness.⁷

4.2 Determining Motive in Flash Attacks

The extreme scale and the rapid, 40-second "hit-and-run" nature of the 22.2 Tbps assault complicate the determination of a singular, simple motive. Several strategic objectives are plausible:

Reconnaissance and Defense Mapping

The most likely primary motive is to test and map the target's defense architecture.³ Simple

disruption could be achieved with far less effort. By executing a record-breaking flash attack, the threat actors force an automated response from the mitigation provider. This action provides valuable intelligence on the provider's capacity thresholds, latency in detection, and the logic governing automated response systems, all of which can be leveraged for future, more complex campaigns designed to evade detection.

Diversiónary Tactics

Hyper-volumetric floods are frequently employed as a sophisticated smokescreen. During the 40 seconds of maximum network saturation, the attack provides cover for a simultaneous, quieter intrusion attempt, such as data exfiltration, lateral network movement, or the deployment of secondary malware.¹⁷ The goal is to maximize network noise and distract security operations centers (SOCs) away from highly targeted, lower-volume probes.

Ransom DDoS (RDoS) Proof of Concept

While many RDoS campaigns are sustained, a record-breaking blast serves as the ultimate "proof of power" demonstration. This capacity can be used to intimidate high-value targets into paying preemptive ransom demands. Ransom DDoS attacks saw a 68% QoQ increase in the second quarter of 2025, confirming that financial extortion is a rapidly growing primary motive.⁶

The sheer size of the Aisuru botnet and its established capability to rapidly pivot between motives (ransom, reconnaissance, diversion) and attack protocols confirm that network security strategy must shift from merely reacting to known signatures (e.g., specific DNS flood vectors) to proactively neutralizing entire threat actors based on their observed behavior and threat intelligence, regardless of the precise protocol they choose for their next assault.

V. Impact Assessment and Economic Fallout

The successful autonomous mitigation of the 22.2 Tbps attack prevented a catastrophic disruption, but analysis of the potential impact provides a crucial validation for comprehensive defense spending.

5.1 Infrastructure Stress Test and Failure Mode

For a critical network infrastructure company relying solely on traditional on-premises appliances, the impact of a 22.2 Tbps flood would have been immediate and total. The network would have suffered immediate **link saturation**, completely overwhelming all available ingress bandwidth and rendering services inaccessible.⁴

Critically, the 10.6 Bpps rate, combined with the carpet bombing across 47,000 ports, would have instantly and completely **exhausted the connection tracking tables and CPU resources** of any stateful on-premises security device (firewalls, deep packet inspection systems). This forced resource starvation would have resulted in critical system failures, potential kernel panics, and multi-hour downtime required for system recovery, network cleanup, and service restoration.¹²

5.2 Realized Impact vs. Potential Catastrophe

The attack was entirely blocked by the mitigation provider's autonomous systems.⁴ The hyper-volumetric traffic was scrubbed at the edge, leveraging the global network capacity to ensure that the malicious packets never reached the target's origin infrastructure.⁴ This outcome is a definitive validation of automated, machine-driven, cloud-based defense architectures.

Had the attack succeeded, the service unavailability would have crippled millions of downstream commercial and consumer clients reliant on the European network infrastructure company. Such an outage would represent a catastrophic failure of critical national infrastructure, leading to severe financial losses, high customer churn, and long-term reputational damage.³ The successful outcome confirms that DDoS defense is no longer a tertiary security concern but a core business continuity requirement.

5.3 Quantification of Downtime Cost and Asymmetry

The economic asymmetry between the cost of launching a DDoS attack and the cost of absorbing its impact is the fundamental factor driving the proliferation of these large-scale

threats. The average cost of a successful, damaging DDoS attack to an enterprise is estimated to hover around **\$500,000** per incident.¹⁸ For unprotected organizations, the financial toll can exceed \$1 million for large companies, with each minute of service downtime costing upwards of **\$6,000**.²⁰

Conversely, the financial barrier to entry for threat actors is extremely low. Attackers can rent the required capacity, often facilitated through DDoS-for-hire services using massive botnets like Aisuru, for as little as **US\$5 to \$7 per hour**.²¹ This stark asymmetry—where minimal investment yields catastrophic potential loss—fundamentally justifies the investment in scaled, external mitigation services as a necessary form of risk avoidance.

Table 1 provides a clear comparative analysis of this economic disparity, reinforcing the strategic value of comprehensive protection.

Table 1: Economic Asymmetry of DDoS Attacks (2025 Context)

Metric	Attacker Cost (Botnet Rental)	Victim Cost (Average Successful Attack)	Downtime Cost per Minute
Financial Outlay (Per Hour)	US\$5 – US\$7 ²¹	N/A	US\$6,000 (for unprotected organizations) ²⁰
Total Loss (Typical Incident)	Near Zero	~\$486,000 – \$500,000 ¹⁹	N/A
Strategic Goal	Maximum Disruption/Reconnaissance	Business Continuity and Reputation Preservation	N/A

VI. Strategic Resilience: Mitigating Terabit-Scale Threats

The 22.2 Tbps attack provides an operational mandate for network security leadership: defense must be globally distributed, autonomous, and multi-layered to address both volumetric and packet-intensive vectors.

6.1 The Imperative of Automation and Edge Capacity

To successfully withstand a 22.2 Tbps traffic flood, organizations must move beyond traditional on-premises appliances. The singular necessary component for volumetric resilience is the leveraging of cloud-based DDoS protection services (DDoS-as-a-Service) that possess massive, globally distributed scrubbing centers with virtually limitless capacity.¹

Given the "flash" nature of these attacks (40 seconds), detection and response must be entirely automated, operating at machine speed.⁴ This requires continuous traffic analysis utilizing advanced Artificial Intelligence (AI) and Machine Learning (ML) algorithms to instantly distinguish malicious traffic from legitimate spikes, enabling instantaneous mitigation. Furthermore, seamless integration between internal infrastructure monitoring tools and external cloud scrubbing centers is paramount, ensuring that attack traffic can be redirected instantly to the edge for cleaning upon detection, as highlighted by successful integration practices.¹²

6.2 Multi-Layered Defense Architecture (L3/L4 Focus)

Effective defense against hyper-volumetric, carpet-bomb threats requires a multi-faceted approach addressing the attack across multiple network layers simultaneously.

A. Network Filtering and Source Protection (External)

Upstream defense is the first line of engagement. Organizations must work proactively with their Internet Service Providers (ISPs) to implement Best Current Practice 38 (BCP38) source IP verification. This technique filters traffic to prevent packets with spoofed source IP addresses from entering the network¹, which is critical for defending against common reflection and amplification vectors, such as DNS flood attacks (which were the top L3/L4 attack vector in Q2 2025).⁶ Although the 22.2 Tbps attack used non-spoofed IPs, BCP38

remains a fundamental component of network hygiene. Additionally, robust network boundaries must employ Access Control Lists (ACLs) and implement aggressive rate limiting to restrict suspicious connection attempts based on protocol, port, or source IP, protecting internal resources from exhaustion.²³

B. Protocol and Anomaly Detection (Internal)

Deployment of real-time traffic monitoring and behavioral analysis tools is essential to detect the nuanced patterns indicative of a carpet bomb—specifically, the simultaneous spike in traffic directed at a distributed range of thousands of ports on a single IP address.²² Security architecture must also be continuously reinforced by integrating real-time threat intelligence feeds. These feeds allow security systems to immediately incriminate and block known botnet sources, such as those associated with Aisuru, ensuring that if a known actor targets one customer, the entire network is immediately protected against similar tactics, regardless of the specific protocol used.⁶

C. Infrastructure Hardening

Physical and virtual infrastructure must be architected for resilience. This includes ensuring redundant network paths and geographically dispersed infrastructure to distribute the attack surface and absorb localized spikes.¹ Protocol hardening is also key: unnecessary network services and protocols should be disabled to minimize the attack surface available for abuse, particularly given the observed flexibility of threat actors to pivot between UDP and TCP attacks.¹³

Table 2 synthesizes the required architectural approach to achieve strategic resilience against current hyper-volumetric threats.

Table 2: Strategic Framework for Hyper-Volumetric Mitigation (L3/L4)

Defense Horizon	Mechanism/Tool	Primary Objective	Relevance to 22.2 Tbps Assault
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Horizon 1: Edge Absorption (L3/L4)	Cloud-based Scrubbing Services	Absorb massive volumetric floods (Tbps scale) globally, close to source	Provides essential capacity against the 22.2 Tbps volume, far surpassing origin network limits ³
Horizon 2: Intelligent Filtering (L3/L4)	AI/ML Anomaly Detection & Threat Intelligence	Identify unique attack signatures (e.g., 10.6 Bpps rate, carpet bomb port spread) and incriminate botnets instantly	Crucial for mitigating the 40-second "hit-and-run" speed and the sophisticated port dispersion used to bypass stateful inspection ⁶
Horizon 3: Origin Hardening (L4/L7)	Rate Limiting, Load Balancing, Internal Firewalls	Prevent resource exhaustion on exposed services and increase system redundancy	Necessary defense against combined multi-vector attacks and maintaining system resilience when the edge is under peak load ⁷
Out-of-Band Defense	BCP38/Source IP Verification	Prevent network reflection/amplification attacks that rely on source IP spoofing	Fundamental network hygiene, addressing common L3/L4 vectors like DNS floods that frequently complement hyper-volumetric assaults ⁶

6.3 Supply Chain Defense (Botnet Source Remediation)

The record-breaking scale of the 22.2 Tbps attack was enabled by the proliferation of compromised IoT devices.⁸ A crucial long-term strategy involves reducing the available infrastructure for botnets like Aisuru. Organizations must advocate for better security practices among their users and vendors: mandatory **firmware updates, immediate retirement of End-of-Life (EoL) devices, replacement of default administrative credentials, and disabling remote administration (WAN access)** unless absolutely necessary.¹⁵ This focus on upstream device security reduces the total pool of compromised nodes, thereby degrading the attacker's ability to launch future hyper-volumetric assaults.

VII. Conclusion: Forecasting the Next Threshold

The successful mitigation of the 22.2 Tbps hyper-volumetric assault in September 2025 is a definitive indicator that the DDoS arms race has entered an era defined by extreme scale, rapid escalation, and machine-speed tactics.

The benchmark established by this event is temporary; the capacity planning assumption for network security must now assume near-limitless absorption capability. The threat evolution confirms that L3/L4 attacks are moving beyond simple bandwidth saturation (Tbps) toward tactical sophistication (Carpet Bombing), explicitly targeting network hardware exhaustion (Bpps). This requires defense systems to focus on processing speed and packet density analysis.

The flexibility demonstrated by the suspected Aisuru botnet, including its rapid pivot to a suspected 29.69 Tbps TCP-based carpet bomb in October 2025¹⁶, strongly suggests that future assaults will be omni-protocol, multi-vector campaigns designed to probe defenses across every layer simultaneously. Successful DDoS resilience is inextricably linked to a holistic strategy that integrates high-capacity, automated cloud scrubbing at the edge with advanced, AI-driven behavioral filtering and proactive measures to reduce the global vulnerability of IoT devices fueling these attacks. For critical infrastructure, this defense paradigm is the singular guarantee of business continuity in the face of escalating digital conflict.

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